

Problem

Design an LC ladder network to realize a low-pass filter with transfer function

$$H(s) = \frac{1}{s^4 + 2.613s^2 + 3.414s^2 + 2.613s + 1}$$

Step-by-step solution

Step 1 of 6

Consider the following low pass filter transfer function.

$$H(s) = \frac{1}{s^4 + 2.613s^2 + 3.414s^2 + 2.613s + 1}$$

Note: The s^2 term of magnitude 2.613 in the numerator of the transfer function is assumed as s^3 , since the results for the available transfer function are not possible.

Therefore,

$$H(s) = \frac{1}{s^4 + 2.613s^3 + 3.414s^2 + 2.613s + 1}$$

The order of the filter is 4.

The following is the LC ladder network for the low pass filter of order 4:

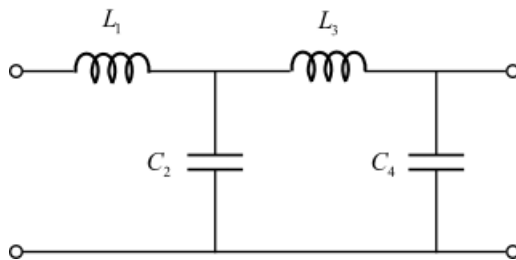


Figure 1

Step 2 of 6

Determine the values of the inductors and capacitors in the circuit of Figure 1.

Group the terms in the denominator of the transfer function into odd and even parts:

$$\begin{aligned} H(s) &= \frac{1}{s^4 + 2.613s^3 + 3.414s^2 + 2.613s + 1} \\ &= \frac{1}{(s^4 + 3.414s^2 + 1) + (2.613s^3 + 2.613s)} \\ &= \frac{1}{1 + \frac{2.613s^3 + 2.613s}{s^4 + 3.414s^2 + 1}} \end{aligned}$$

The following is the expression for the transfer function of a low pass filter when $Y_L = 1$:

$$H(s) = \frac{-Y_{21}}{1 + Y_{22}}$$

Therefore,

$$\begin{aligned} Y_{21} &= \frac{-1}{2.613s^3 + 2.613s} \\ Y_{22} &= \frac{s^4 + 3.414s^2 + 1}{2.613s^3 + 2.613s} \end{aligned}$$

Step 3 of 6

The output driving-point (input port short-circuited) admittance of the network shown in Figure 1 is Y_{22} . So, the realization of Y_{22} automatically realize Y_{21} .

Determine the values of L and C that results the admittance Y_{22} .

Draw the LC network by short circuiting the input port:

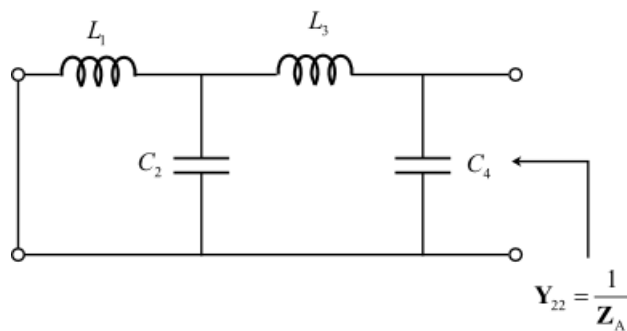


Figure 2

Step 4 of 6

The simplified expression for $\mathbf{Y}_{22}$ equals $sC_4 + \mathbf{Y}_A$.

Simplify the fraction $\mathbf{Y}_{22}$ using long division:

$$\frac{2.613s^3 + 2.613s}{s^4 + s^2} \Big) s^4 + 3.414s^2 + 1 \Big(0.383s$$

$$\frac{2.414s^2 + 1}{2.414s^2 + 1}$$

The following is the simplified expression for $\mathbf{Y}_{22}$.

$$\mathbf{Y}_{22} = 0.383s + \frac{2.414s^2 + 1}{2.613s^3 + 2.613s}$$

Therefore,

$$C_4 = 0.383 \text{ F}$$

$$\mathbf{Y}_A = \frac{2.414s^2 + 1}{2.613s^3 + 2.613s}$$

The following is the impedance corresponds to $\mathbf{Y}_A$.

$$\mathbf{Z}_A = \frac{1}{\mathbf{Y}_A}$$

$$= \frac{1}{\left(\frac{2.414s^2 + 1}{2.613s^3 + 2.613s} \right)}$$

$$= \frac{2.613s^3 + 2.613s}{2.414s^2 + 1}$$

The simplified expression for $\mathbf{Z}_A$ equals $sL_3 + \mathbf{Z}_B$.

Simplify the fraction $\mathbf{Z}_A$ using long division:

$$\frac{2.414s^2 + 1}{2.613s^3 + 2.613s} \Big) 2.613s^3 + 2.613s \Big(1.08s$$

$$\frac{2.613s^3 + 1.08s}{1.53s}$$

The following is the simplified expression for $\mathbf{Z}_A$.

$$\mathbf{Z}_A = 1.08s + \frac{1.53s}{2.414s^2 + 1}$$

Therefore,

$$L_3 = 1.08 \text{ H}$$

$$\mathbf{Z}_B = \frac{1.53s}{2.414s^2 + 1}$$

The following is the admittance corresponds to $\mathbf{Z}_B$.

$$\mathbf{Y}_B = \frac{1}{\mathbf{Z}_B}$$

$$= \frac{1}{\left(\frac{1.53s}{2.414s^2 + 1} \right)}$$

$$= \frac{2.414s^2 + 1}{1.53s}$$

The simplified expression for $\mathbf{Y}_B$ equals $sC_2 + \frac{1}{sL_1}$.

Simplify the fraction $\mathbf{Y}_B$ using long division:

$$\begin{array}{r} 1.53s \overline{) 2.414s^2 + 1(1.578s} \\ \underline{2.414s^2} \\ 1 \end{array}$$

The following is the simplified expression for $\mathbf{Y}_B$.

$$\mathbf{Y}_B = 1.578s + \frac{1}{1.53s}$$

Therefore,

$$C_2 = 1.578 \text{ F}$$

$$L_1 = 1.53 \text{ H}$$

Step 6 of 6

Draw the LC ladder network corresponds to the transfer function $H(s)$:

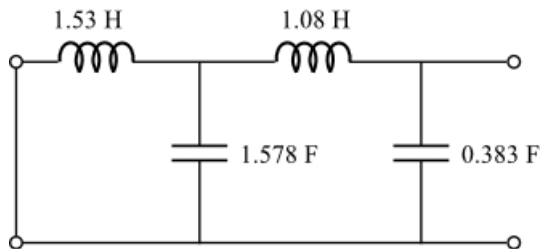


Figure 3